

Practice Exercise Media Inheritance

Design and Implement a Java program as follows:

- 1) Model *Media* object to include at minimum the following:
 - **Member variables (attributes):** id (integer), title (string), year published (integer)
 - **Functionality:** constructor with all the attribute values, calculate rental fee (\$3.50 flat fee), change title, change year published, and get each attribute value.
 - The class should be declared abstract as it represents a generic media and we do not want to allow a creation of this class directly
 - Note that id value cannot be changed after creation.
- 2) Model *EBook* object which is a child of *Media* and includes:
 - **Member variables (attributes):** inherits from parent plus attribute for number of chapters (integer)
 - **Functionality:** inherited functionality plus constructor with all the attribute values, get and set methods for local attribute, display values to the console
 - The rental fee for *EBook* is the number of chapters * 0.10 + \$1 if *EBook* was published this year
- 3) Model *MusicCD* object which is a child of *Media* and includes:
 - **Member variables (attributes):** inherits from parent plus attribute for length in minutes (integer)
 - **Functionality:** inherited functionality plus constructor with all the attribute values, get and set methods for local attribute, display values to the console
 - The rental fee for *MusicCD* is the length * 0.02 + \$1 if *MusicCD* was published this year
- 4) Model *MovieDVD* object which is a child of *Media* and includes:
 - **Member variables (attributes):** inherits from parent plus attribute for size in megabytes (double)
 - **Functionality:** inherited functionality plus constructor with all the attribute values; get and set methods for local attribute, display values to the console
 - The rental fee for *MovieDVD* is the flat fee \$3.50
- 5) Create *RunRental* class with main method which does the following:
 - Creates an instance of each type of media
 - Display each instance's attribute data as well as the rental fee by calling it's calculate method
 - Change *EBook* instance's number of chapters and display new rental fee
 - Change *MusicCD* instance's length and year and display new rental fee
- 6) Make sure you implement correct encapsulation (private attributes, get methods, and set methods when value can change)
- 7) To get current year you can use: `Calendar.getInstance().get(Calendar.YEAR);`

SAMPLE RUN

```
EBook [ id=123, title=Forever Young, year=2018, chapters=20]    Rental fee=$2.00
```

MusicCD [id=124, title=Beyond Today, year=2020, length=114min] Rental fee=\$3.28
MovieDVD [id=125, title=After Tomorrow, year=2020, size=120.0MB] Rental fee=\$3.50

Changing EBook chapters to 25: EBook [id=123, title=Forever Young, year=2018,
chapters=25]
New Rental fee=\$2.50

Changing MusicCD length to 120: MusicCD [id=124, title=Beyond Today, year=2020,
length=120min]
New Rental fee=\$3.40

Sample Solution:

```
/*
 * abstract class to represent a generic media
 */
public abstract class Media {

    // attributes
    private int id;
    private String title;
    private int year; // validate that 4 digits

    // constructor
    public Media(int id, String title, int year) {

        this.id = id;
        this.title = title;
        this.year = year;
    }

    // get methods
    public int getId() {
        return id;
    }

    public String getTitle() {
        return title;
    }

    public int getYear() {
        return year;
    }

    // set methods
    public void setTitle(String title) {
        this.title = title;
    }

    public void setYear(int year) {
        this.year = year;
    }

    // calculate rental fee; for generic media it is flat fee $3.50
    public double calculateRentalFee() {
        return 3.50;
    }
}

import java.util.Calendar;

public class EBook extends Media {

    // local attributes
    private int numChapters;
```

```

// constructor
public EBook(int id, String title, int year, int chapters) {
    super(id, title, year);
    numChapters = chapters;
}

// get method
public int getNumChapters() {
    return numChapters;
}

// set method
public void setNumChapters(int numChapters) {
    this.numChapters = numChapters;
}

// override parent's
@Override
public double calculateRentalFee() {
    double fee = numChapters * 0.10; // basic fee
    int currYear = Calendar.getInstance().get(Calendar.YEAR);

    if (this.getYear() == currYear)
        fee += 1; // add $1.00 fee

    return fee;
}

@Override
public String toString() {
    return "EBook [ id=" + getId() + ", title=" + getTitle()
        + ", year=" + getYear() + ", chapters=" + numChapters +
    " ]";
}

}

public class MovieDVD extends Media {

    // local attributes
    private double size; // value in MB

    // constructor
    public MovieDVD(int id, String title, int year, double size) {
        super(id, title, year);
        this.size = size;
    }

    // get method
    public double getSize() {
        return size;
    }

    // set method
    public void setSize(double size) {

```

```

        this.size = size;
    }

    // inherits calculate rental fee method and no different calculation so should
    not override

    @Override
    public String toString() {
        return "MovieDVD [ id=" + getId() + ", title=" + getTitle() + ", year="
            + getYear() + ", size=" + size + "MB]";
    }
}

import java.util.Calendar;

public class MusicCD extends Media {

    // local attributes
    private int length;

    // constructor
    public MusicCD(int id, String title, int year, int length) {
        super(id, title, year);
        this.length = length;
    }

    // get method
    public int getLength() {
        return length;
    }

    // set method
    public void setLength(int length) {
        this.length = length;
    }

    // override parent's
    @Override
    public double calculateRentalFee() {
        double fee = length * 0.02; // basic fee
        int currYear = Calendar.getInstance().get(Calendar.YEAR);

        if (this.getYear() == currYear)
            fee += 1; // add $1.00 fee

        return fee;
    }

    @Override
    public String toString() {
        return "MusicCD [id=" + getId() + ", title=" + getTitle() + ", year="
            + getYear() + ", length=" + length + "min]";
    }
}

public class RunRental {

```

```

public static void main(String[] args) {

    // create instances of the ebook and display
    EBook ebook = new EBook(123, "Forever Young", 2018, 20);
    System.out.print(ebook.toString());
    System.out.printf("    Rental fee=$%.2f\n", ebook.calculateRentalFee());

    // create instances of the music cd and display
    MusicCD cd = new MusicCD(124, "Beyond Today", 2020, 114);
    System.out.print(cd.toString());
    System.out.printf("    Rental fee=$%.2f\n", cd.calculateRentalFee());

    // create instances of the movie dvd and display
    MovieDVD dvd = new MovieDVD(125, "After Tomorrow", 2020, 120);
    System.out.print(dvd.toString());
    System.out.printf("    Rental fee=$%.2f\n", dvd.calculateRentalFee());

    ebook.setNumChapters(25);
    System.out.print("\nChanging EBook chapters to 25:    ");
    System.out.println(ebook.toString());
    System.out.printf("    New Rental fee=$%.2f\n",
ebook.calculateRentalFee());

    cd.setLength(120);
    System.out.print("\nChanging MusicCD length to 120:    ");
    System.out.println(cd.toString());
    System.out.printf("    New Rental fee=$%.2f\n",
cd.calculateRentalFee());
}

}

```